Comparing forces is about more than just a strength test, though. The idea that all the forces can somehow be reframed as different aspects of the same thing, in extremely high-energy environments, is generally considered to be key to truly understanding how physics works. We live in hope that there's some ultimate theory out there—a theory of everything—that unites all the forces of particle physics and gravity to explain, well, everything.

But so far, gravity won't play along. We have a rock-solid theory of the electroweak force (a unification of electromagnetism and the weak nuclear force), confirmed by experiments. We also have some perfectly promising leads on a Grand Unified Theory uniting the electroweak and strong nuclear force. But every time we try to bring gravity in, its feebleness ruins the whole picture. Even aside from that, gravity and quantum mechanics (which describes the workings of all the other forces) explicitly clash in their predictions about things like what should happen at the edge of a black hole. Finding a way to bring gravity into line would help immensely.

So there seem to be a few options here. One obvious one is to abandon the whole idea of unification, and just let gravity be its own special snowflake of a theory, unconnected to the rest of physics. It's totally possible that there's no theory of everything, and we're never going to piece it all together in any sensible way. But just typing that makes my physicist toes curl, so maybe we can set that to the side for now in the "BREAK GLASS IN CASE OF EXISTENTIAL EMERGENCY" cabinet.

A much more appealing and intellectually exciting idea is that the problem is our theory of gravity: general relativity needs to be altered or replaced, and when that happens, it'll all fit together. There's been no shortage of impressive, well-motivated attempts in this direction. Quantum gravity theories, of which string theory and loop quantum gravity are the most famous examples, continue to be hot topics among theorists trying to find a way to unite particle physics with gravity and tie it all up with string. Or loops. You get the idea. In each of these scenarios, you end up with a gravity theory that can be *quantized*—expressed in terms of particles and fields rather than forces or spatial curvature—and these particles and fields mesh nicely with those of the quantum field theories that explain interactions between quarks and electrons and photons and the whole subatomic world. In this picture, gravitational forces would be manifestations of the exchange of particles called gravitons, just like an electric field is due to photons moving between objects. And gravitational